

# Sandipan Chakraborty

Backend Engineer • AI & LLM Systems • Distributed Architecture

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## Professional Summary

Backend & AI Systems Engineer with experience designing and deploying *real-time, low-latency AI pipelines, LLM-powered applications, and distributed backend systems.*

Designed and deployed **production-grade systems**, including asynchronous pipelines processing **1K–3K events/day** (scalable to 10K+) and voice AI systems with **~120–180ms latency** (p95 <250ms). Strong focus on **system design, fault-tolerant architecture, and end-to-end AI pipelines** (input → processing → decision → response).

## EXPERIENCE

### AI & Automation Intern (Paid)

IndiiServe Solutions Pvt. Ltd., Kolkata

Feb 2026 – Present

- **Architected production voice AI system** (FastAPI + WebSockets + AWS Bedrock) achieving ~120–180ms latency (p95 <250ms), handling concurrent real-time requests via Exotel telephony integration
- **Designed and owned end-to-end system architecture** for real-time AI pipeline (call → inference → response), including service decomposition, API contracts, and multi-tenant AWS EC2 deployment
- **Engineered OIE async pipeline** processing 1K–3K events/day (scalable to 10K+) and WhatsApp Recovery System (Redis + Celery) with idempotent retries and Dead-Letter Queue, reducing delivery failures to near-zero
- **Implemented safety-critical routing system** for emergency escalation and developed tool-driven LLM architecture for structured retrieval
- Built **OCR-based data extraction pipelines** converting unstructured documents into structured, queryable records
- Deployed and maintained services on AWS EC2, improving reliability, fault tolerance, and performance

### Machine Learning Intern -

LABMENTIX (Remote)

Aug 2025 – Oct 2025

- Developed and deployed 3 RESTful APIs (FastAPI) for ML model serving, cutting manual data-processing time by ~40% across data workflows.
- Trained and evaluated classification & regression models; applied feature engineering and cross-validation, improving prediction accuracy by ~15% over baseline.

## PROJECTS

### U-MEM: Universal Document Intelligence & Memory System

**Tech:** Python, Sentence Transformers, FAISS, OCR, SHA-256, Git

([GITHUB](#))

- Designed **semantic retrieval pipeline** (embedding → HDBSCAN → FAISS) enabling sub-second document search
- Built end-to-end ingestion, OCR, embedding, clustering, and retrieval pipeline for scalable knowledge access

### Cloud Cost Knowledge Graph

**Tech:** FastAPI, Neo4j (Graph Database), Streamlit, Sentence Transformers, Scikit-learn

([GITHUB](#))

- Constructed knowledge graph with **8.5K+ nodes and 43K+ relationships** modeling cloud service dependencies
- Implemented **hybrid retrieval (vector + graph + Cypher queries)** with LLM-assisted cost optimization insights

## TECHNICAL SKILLS

**Languages:** Python, C++

**Backend & Systems:** FastAPI, REST APIs, WebSockets, Async I/O, Event-Driven Architecture, Microservices, Streaming, Distributed Systems

**AI / LLM:** LLM Integration, Prompt Engineering, RAG Pipelines, Retrieval-Augmented Generation, Semantic Search, FAISS, Sentence Transformers, OCR, AWS Bedrock (Claude 3.5 Sonnet, Nova Sonic)

**Infrastructure:** AWS EC2, Docker, Redis, Celery, Background Jobs, Fault Tolerance, Retry Mechanisms

**Databases:** PostgreSQL, Neo4j (Graph Database), SQLite

## Soft SKILLS

Analytical Thinking • Communication • Team Collaboration • Adaptability • Research Mindset

## EDUCATION

**Techno International Newtown**

B.Tech – Computer Science & Business Systems | CGPA: 7.5

Kolkata, WB

Aug 2022 – May 2026